ONLINE QUIZ SYSTEM

Software Test Plan

Course: Software Engineering
Class: 5th Semester – Section H
Team: 13
Document ID: OQS-TP-02
Version: 1.0
Date: 28th Sept 2026

Team Members

Name	SRN
Shwetha S	PES1UG24CS452
Srishti Rao	PES1UG24CS470
Pratistha Majumder	PES1UG25CS831
Sharvari Naik	PES1UG24CS431

Document Control

Version	Date	Description
1.0	02-11-2026	Initial consolidated Software Test Plan

1. Introduction

1.1 Purpose

The purpose of this Software Test Plan is to define the testing activities, test strategy, test environment, test cases and validation methods for the Online Quiz System.

The plan ensures that the developed application is tested against the requirements specified in OQS-SRS-01.

1.2 Scope

Testing covers:

- Participant registration/details entry and quiz access.
- Question display and navigation.
- Answer submission and score calculation.
- Result display.

- Administrator authentication.
- Question management.
- Input validation.
- Security and access control.
- Performance, usability, compatibility and reliability.

Advanced proctoring, payment processing and AI-based question generation are outside the current testing scope.

1.3 References

1. Online Quiz System – Software Requirements Specification, OQS-SRS-01, Version 1.0.
2. Software Engineering project guidelines.
3. Online Quiz System project requirements.

1.4 Test Items

The following components shall be tested:

- HTML, CSS and JavaScript frontend.
- Node.js and Express.js backend.
- MySQL database.
- REST API endpoints.
- Integrated Online Quiz System.

2. Test Objectives and Strategy

2.1 Test Objectives

1. Verify that all functional requirements work correctly.
2. Verify that the system meets its non-functional requirements.
3. Validate administrator authentication and access control.
4. Verify correct calculation and display of quiz results.
5. Identify defects before project demonstration and submission.
6. Confirm that invalid inputs and unexpected errors are handled safely.

2.2 Testing Levels

Unit Testing: Individual functions such as input validation, authentication and score calculation are tested independently.

Integration Testing: Communication between the frontend, backend and MySQL database is tested.

System Testing: The complete application is tested from the participant's and administrator's perspectives.

Acceptance Testing: The team and guide review whether the application satisfies the agreed project requirements.

2.3 Testing Types

- Functional testing
- Non-functional testing
- Security testing
- Performance testing
- Usability testing
- Compatibility testing
- Input validation testing
- Regression testing

2.4 Testing Approach

Both positive and negative test scenarios shall be used.

Positive testing verifies that valid inputs produce the expected results.

Negative testing verifies that invalid inputs, unauthorised requests and unexpected conditions are handled correctly.

Testing shall be carried out in a separate test environment using sample data.

3. Test Environment

3.1 Hardware Requirements

The test environment shall use a computer capable of running the frontend, backend server and MySQL database.

The actual processor, memory and operating system details shall be recorded during test execution.

3.2 Software Requirements

Component	**Technology**
Frontend	HTML, CSS, JavaScript
Backend	Node.js
Framework	Express.js
Database	MySQL
Browser	Chrome, Edge or Firefox
API Testing	Postman or equivalent
Code Editor	Visual Studio Code
Version Control	Git and GitHub

3.3 Test Data

The team shall prepare:

- Valid administrator credentials.
- Invalid administrator credentials.
- Sample quiz questions.
- Correct and incorrect answer combinations.
- Valid and invalid participant details.
- Questions with missing or invalid answer options.
- Sample participant attempts.

Only test data shall be used during testing.

4. Test Management

4.1 Roles and Responsibilities

All team members shall participate in reviewing the test cases, executing assigned tests, recording defects and verifying corrections.

Each test case shall be assigned to a team member before execution.

4.2 Entry Criteria

Testing may begin when:

1. The required feature has been implemented.
2. The application can be started in the test environment.
3. The MySQL database and test data are available.
4. The relevant test cases have been reviewed.
5. Required test accounts are configured.

4.3 Exit Criteria

Testing may be considered complete when:

1. All planned test cases have been executed or formally deferred.
2. Failed cases have corresponding defect records.
3. Critical defects have been resolved or formally accepted as outstanding.
4. Regression testing has been completed for corrected defects.
5. Test results and evidence have been documented.

4.4 Defect Reporting

Each defect shall include:

- Defect ID.
- Related requirement and test case ID.
- Description of the problem.
- Steps to reproduce.
- Expected result.
- Actual result.
- Severity.
- Assigned team member.
- Current status.

5. Test Design and Validation

5.1 Security Validation

Security validation verifies that the system protects administrator credentials, restricts administrative functions, validates inputs and prevents disclosure of sensitive information.

The following security test cases shall be performed in the authorised test environment.

Test ID	Requirement	Test Scenario	Expected Result
TC-SEC-01	FR-08, SEC-01, NFR-04	Access an administrator API without logging in.	Request is rejected with an appropriate unauthorised response.
TC-SEC-02	SEC-01, SEC-06	Attempt question management using a participant account.	Access is denied and no database changes occur.
TC-SEC-03	NFR-03, SEC-02	Inspect the stored administrator credential.	Only a salted password hash is stored; plaintext password is not exposed.
TC-SEC-04	FR-13, SEC-03	Submit SQL injection-like input in a question field.	Input is safely handled or rejected; no unintended database operation occurs.
TC-SEC-05	SEC-04	Inspect the deployed authentication communication and session configuration.	Credentials and session identifiers are protected using secure configuration.

TC-SEC-06	NFR-10, SEC-05	Trigger a controlled server error.	The client receives a generic error message without stack traces or database details.

5.2 Functional Testing

Test ID	Requirement	Test Scenario	Expected Result
TC-F-01	FR-01	Open the quiz start page.	Quiz information and start option are displayed.
TC-F-02	FR-02, FR-13	Submit participant details with a required field missing.	System displays a validation message and prevents starting the quiz.
TC-F-03	FR-03	Start a quiz containing questions and answer options.	Questions and available options are displayed correctly.
TC-F-04	FR-04, NFR-05	Select an answer, navigate to another question and return.	Previously selected answer remains available.
TC-F-05	FR-05	Submit an active quiz attempt.	System records the submission and prevents unintended duplicate submission.
TC-F-06	FR-06	Submit a quiz containing known correct and incorrect answers.	The server calculates the expected score correctly.
TC-F-07	FR-07	View the result after submission.	Correct result information is displayed.
TC-F-08	FR-08, SEC-01	Log in using valid and invalid administrator credentials.	Valid credentials grant access; invalid credentials are rejected.
TC-F-09	FR-09, FR-13	Add a valid question as an administrator.	The question is saved and appears in the question list.
TC-F-10	FR-10	Edit an existing question.	Updated question details are saved and displayed.
TC-F-11	FR-11	Delete a question after confirmation.	The question is deleted or deactivated according to the agreed policy.
TC-F-12	FR-12	Open the question list as an administrator.	The administrator can view available questions.
TC-F-13	FR-13, NFR-09	Submit a question with missing or invalid answer configuration.	The system rejects the invalid question without saving incomplete data.

5.3 Non-Functional Testing

Test ID	Requirement	Test Scenario	Expected Result
TC-N-01	NFR-01	Measure page and API response times under normal conditions.	At least 95% of measured requests complete within the proposed 3-second target.
TC-N-02	NFR-02	Simulate 50 concurrent participant sessions.	Application remains usable and no data corruption occurs.
TC-N-03	NFR-06	Review interface labels, readability and navigation.	Main operations are understandable and validation messages are clear.
TC-N-04	NFR-07	Test the application in Chrome, Edge and Firefox.	Core workflows operate correctly in the supported browsers.
TC-N-05	NFR-08, NFR-09	Review module organisation and database consistency.	Modules are separated and invalid data does not corrupt records.
TC-N-06	NFR-10	Simulate temporary database unavailability.	The application displays a safe error and recovers after the database is restored.

5.4 Input Validation Testing

The system shall be tested using:

1. Empty required fields.
2. Invalid participant details.
3. Missing question text.
4. Missing answer options.
5. Invalid question identifiers.
6. Invalid answer selections.
7. Unexpected data types.
8. Excessively long input values.

Expected result: Invalid data shall be rejected by server-side validation and shall not be stored in the database.

5.5 Usability Testing

The participant and administrator interfaces shall be reviewed for:

- Clear labels and instructions.
- Readable question and option text.
- Understandable navigation.
- Visible submit and confirmation controls.
- Clear error and success messages.
- Consistent page layout.

5.6 Regression Testing

Whenever a defect is corrected, related test cases shall be executed again to ensure that the correction has not affected existing functionality.

6. Requirement Traceability Matrix

Requirement ID	Related Test Cases
FR-01	TC-F-01
FR-02	TC-F-02
FR-03	TC-F-03
FR-04	TC-F-04
FR-05	TC-F-05
FR-06	TC-F-06
FR-07	TC-F-07
FR-08	TC-F-08, TC-SEC-01
FR-09	TC-F-09
FR-10	TC-F-10
FR-11	TC-F-11
FR-12	TC-F-12
FR-13	TC-F-02, TC-F-09, TC-F-13, TC-SEC-04
NFR-01	TC-N-01
NFR-02	TC-N-02
NFR-03	TC-SEC-03

NFR-04	TC-SEC-01
NFR-05	TC-F-04
NFR-06	TC-N-03
NFR-07	TC-N-04
NFR-08	TC-N-05
NFR-09	TC-F-13, TC-N-05
NFR-10	TC-N-06, TC-SEC-06
SEC-01	TC-SEC-01, TC-SEC-02
SEC-02	TC-SEC-03
SEC-03	TC-SEC-04
SEC-04	TC-SEC-05
SEC-05	TC-SEC-06
SEC-06	TC-SEC-02

7. Test Execution and Reporting

7.1 Test Execution Record

For every test case, the team shall record:

Field	Description
Test Case ID	Unique test identifier
Tester	Name of team member
Execution Date	Date of execution
Actual Result	Observed system behaviour
Status	Pass / Fail / Blocked / Not Run
Defect ID	Related defect, if any
Evidence	Screenshot, log or other proof

7.2 Test Status

- **Not Run:** Test has not yet been executed.
- **Pass:** Actual result matches expected result.
- **Fail:** Actual result differs from expected result.

- **Blocked:** Test cannot be executed because of a dependency or defect.

All test cases in this initial plan shall remain marked **Not Run** until actual execution.

7.3 Risks and Mitigation

Risk	Mitigation
Incomplete features	Execute tests after the relevant feature is available.
Database connection failures	Verify database configuration before testing.
Insufficient test data	Prepare controlled sample records.
Security vulnerabilities	Perform security validation before release.
Requirement changes	Update the SRS, test cases and traceability matrix together.

8. Conclusion

This Test Plan provides a structured approach for validating the Online Quiz System. It covers functional, non-functional, security, usability and input-validation testing.

The test results shall be recorded after execution and reviewed by the project team before final submission.